



- Q1.** Differentiate between DataFrame and a Series in Pandas and create one of each?
- Q2.** Create a DataFrame from a dictionary containing student names, their ages, and their grades. Print the DataFrame?
- Q3.** Find the shape of the above created DataFrame and display the column names?
- Q4.** Load a CSV file and display the first five rows?
- Q5.** Create a DataFrame using a list of dictionaries where each dictionary represents a person with their name, age, and city. Print the DataFrame?
- Q6.** Using the DataFrame created from the list of dictionaries representing individuals (with their names, ages, and cities), perform the following tasks:
- Create a new Series containing the college IDs for each individual. Assume the college id's to be c001, c002, c00n
 - Set the newly created Series of college IDs as the index of the DataFrame.
 - Print the updated DataFrame to display the college IDs as the index.
- Q7.** Use NumPy to create a 2D array of random numbers and convert it into a DataFrame with specified column names?
- Q8.** Create an empty DataFrame and then perform the following tasks:
- Add a column named "Name" and populate it with a list of names: ["Shiva", "Sunny", "Gayatri", "Pradeep", "Reshmi"].
 - Add a column named "Age" and populate it with a list of ages: [25, 26, 25, 27, 26].
 - Add a column named "Hobby" and populate it with a list of Hobbies: ["Hyderabad", "Chennai", "Kerala", "Hyderabad", "Rajasthan"].
- Q9.** Create a DataFrame that contains some NaN values and demonstrate how to check for the presence of these values.
- Q10.** Given the following data representing products, their prices, and quantities, perform the following tasks:
- Create a DataFrame using the provided lists for "Product", "Price", and "Quantity".
 - Set the "Product" column as the index of the DataFrame.
 - -> Data :
 - "Product": ["A", "B", "C"],
 - "Price": [10.99, 15.49, 7.99],
 - "Quantity": [100, 200, 150]
- Q11.** Given the following data representing products, their prices, and quantities, perform the following tasks:
- Create a DataFrame using the provided lists for "Product", "Price", and "Quantity".
 - Sort the DataFrame based on the **Price** column in ascending order.
 - -> Data :
 - "Product": ["A", "B", "C"],



- "Price": [10.99, 15.49, 7.99],
- "Quantity": [100, 200, 150]

Q12. Create a DataFrame that contains a mix of integers, floats, and strings. Print the data types of each column?

Q13. Create a DataFrame with the following columns: ['name', 'age', 'city']. Use a dictionary to rename **name to first_name**, **age to years**, and **city to location**. Display the DataFrame after renaming and print their datatypes?

Q14. Given a DataFrame with columns ['Product', 'Price', 'Quantity'], rename only the price column to cost. Show the resulting DataFrame without using `inplace` function:

- data:
 - `pd.DataFrame({"Product": ["A", "B", "C"],`
 - `"Price": [10.99, 15.49, 7.99],`
 - `"Quantity": [100, 200, 150]})`

Q15. Load employees csv file into a pandas DataFrame:

- How can you display the first 10 rows of a DataFrame?
- How to filter the employees whose age is less than 30?

Q16. Retrieve information of employees from the IT department.

- List employees who joined after January 1, 2020.

Q17. Calculate the average salary of employees in each department.

- Determine the total number of employees in each department.

Q18. Add a new column `YearsWithCompany` that calculates the years each employee has worked based on their joining date.

- Update salaries by increasing each employee's salary by 5%.

Q19. Sort the employees by salary in descending order.

- Rank employees by age within each department.

Q20. Load the patients dataset into a Pandas DataFrame.

- Check for any missing values in the dataset and handle them (e.g., fill or drop).
- Get a summary of the dataset using `describe()` to understand distributions.

Q21. How many male and female patients are there?

- What is the average cholesterol level among patients with diabetes?

Q22. Find the average heart rate for each medication type.

- Calculate the sum of hospital visits by gender.

Q23. Filter out patients who have high cholesterol levels (e.g., greater than 250)



- Identify patients with both high cholesterol and high blood pressure.

Q24. Rank patients by Cholesterol levels, assigning the highest cholesterol level a rank of 1.

Q25. Display the frequency of Diabetes by Medication.

- Find the average Cholesterol and Blood_Pressure for each Gender.

Q26. Find the correlation between age and cholesterol levels across the entire dataset.

- Determine the gender distribution of patients who are taking Metformin.

Q27. Merge the given two DataFrames into one with common data and overall data:

- 1st DataFrame :
 - "Employee_ID": [101, 102, 103, 104],
 - "Name": ["Alice", "Bob", "Carol", "Dan"],
 - "Department": ["HR", "IT", "Marketing", "IT"]
- 2nd DataFrame :
 - "Employee_ID": [101, 102, 105, 106],
 - "Salary": [50000, 70000, 45000, 40000]

Q28. Identify the employees in employee_details who don't have a corresponding entry in employee_salaries.

- Calculate both the total and average salary per department (including departments with missing salary information).

Q29. Identify employees who have a salary greater than 60,000.

- Add a new column Bonus, where the bonus is calculated as 5% of the salary for HR and Marketing departments and 10% for the IT department.

Q30. Determine which department has the highest average salary?

- Add a new column called Experience_Level, categorizing employees as "Junior" if their salary is below 50,000, "Mid" if between 50,000 and 70,000, and "Senior" if above 70,000?